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Training a machine learning model for hardware component identification

Abstract

A system and a method are disclosed for training a machine-learned model. A device retrieves entries from a database that each correspond a hardware component to a value. The device inputs the entries into a weighting model, and the weighting model outputs weights for the values. The device generates a training set including data formed by pairing each respective hardware component to its respective weighted value, and trains the machine-learned model using the training set. The device receives new data comprising a hardware component and a respective value, determines weights therefor, and re-trains the machine-learned model accordingly. Responsive to detecting a trigger, the device uses the machine-learned model to generate a searchable database, and outputs results to search queries including a value for a queried hardware component and a confidence that the value is correct.

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Background/Summary

TECHNICAL FIELD

(1) The disclosure generally relates to the field of machine learning, and more particularly relates to effective training of machine learning to evaluate hardware components.

BACKGROUND

(2) Aircraft hardware component suppliers and entities responsible for operating and maintaining aircraft do not have access to consensus values corresponding to those hardware components. Existing processes for deriving values are subject to back-and-forth network communications that expend unnecessary bandwidth, both in the needless two-way communications and duplication of communications across multiple entities, and that can take long amounts of time to resolve. Moreover, the existing processes are subject to data entry errors thus resulting in inaccuracy.

SUMMARY

(3) Systems and methods are disclosed herein for training and applying a machine-learned model to determine consensus values for aircraft hardware components. The model is trained in an advantageous manner that avoids a need to maintain large data lakes of historical training data, while still enabling re-training of the model based on new data and/or depreciation of old data. The model is used to generate a searchable database for resolving queries associated with a hardware component. In responding to queries, confidences output by the model may be provided in addition to aircraft hardware component values, thus enabling simple identification of aircraft hardware components that are subject to large variance.

(4) In an embodiment, a device trains a machine-learned model. The training may include

retrieving a plurality of entries from a database, each entry corresponding a respective hardware component to a respective value, and inputting at least a portion of data of each respective entry into a weighting model, the weighting model outputting a weight to be applied to each respective value. The device may then generate a training set, the training set having training data formed by pairing each respective hardware component as a label as paired to their respective values as weighted by their respective weights. The device may finally train the machine-learned model using the training set.

(5) The device may receive new data comprising a hardware component and a respective value, and may generate weighted new data by inputting the new data into the weighting model, the weighting model outputting a weight to be applied to the respective value of the new data. The device re-trains the machine-learned model using the training set and the weighted new data.

(6) Following training and any amount of iteration of re-training, responsive to detecting a trigger, the device generates, using the machine-learned model, a searchable database. The device receives a query from a user comprising an indicated hardware component, searches the searchable database for a result matching the query, and outputs a result including a value for the indicated hardware component and a confidence that the value is correct.

(7) In an embodiment, the device retrieves historical data from a plurality of source databases, the historical data including entries each including a hardware component identifier and an associated value. The device receives new data from a plurality of input databases, the new data comprising a respective hardware component identifier and a respective associated value. The device creates a synthesized set of data by identifying a first subset of data comprising data from the historical data and the new data that is associated with an anomaly, identifying a second subset of data from the historical data and the new data that is associated with an attenuation signal, and identifying a third subset of data from the historical data and the new data that includes data not assigned to the first subset or the second subset.

(8) The device updates the synthesized set of data by discarding or archiving the first subset of data from the synthesized set of data, and by weighting each entry of the second subset of data based on its respective attenuation signal. The device generates a searchable database, the searchable database including each hardware component named by an entry of the synthesized set of data, along with an associated value determined based on the weighted value of the entry. The device receives receiving user input of a search query, and outputs search results based on a comparison of the user input of the search query to entries of the searchable database.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The disclosed embodiments have other advantages and features which will be more readily apparent from the detailed description, the appended claims, and the accompanying figures (or drawings). A brief introduction of the figures is below.

(2) FIG. 1 illustrates one embodiment of a system environment for operating a hardware component service.

(3) FIG. 2 illustrates one embodiment of exemplary modules and databases used by the hardware component service.

(4) FIG. 3 illustrates one embodiment of a user interface showing exemplary manners of searching for hardware components and receiving results.

(5) FIG. 4 is a block diagram illustrating components of an example machine able to read instructions from a machine-readable medium and execute them in a processor (or controller).

(6) FIG. 5 is a flow diagram showing an exemplary process for training a machine learning model to generate a searchable database relating to hardware components.

(7) FIG. 6 is a flow diagram showing another exemplary process for generating a searchable database.

DETAILED DESCRIPTION

(8) The Figures (FIGS.) and the following description relate to preferred embodiments by way of illustration only. It should be noted that from the following discussion, alternative embodiments of the structures and methods disclosed herein will be readily recognized as viable alternatives that may be employed without departing from the principles of what is claimed.

(9) Reference will now be made in detail to several embodiments, examples of which are illustrated in the accompanying figures. It is noted that wherever practicable similar or like reference numbers may be used in the figures and may indicate similar or like functionality. The figures depict embodiments of the disclosed system (or method) for purposes of illustration only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles described herein.

(10) System Environment of Hardware Component Service

(11) FIG. 1 illustrates one embodiment of a system environment for operating a hardware component service. As depicted in FIG. 1, environment **100** includes client device **110** with application **111** installed thereon, network **120**, hardware component service **130**, input database **140**, and source databases **150**. Client device **110** may be any device configured to receive input from a user and communicate with aircraft component service **130**. Exemplary client devices include personal computers, smartphones, tablets, Internet-of-Things (IoT) devices, laptops, kiosks, and so on. Communications between client device **110** and hardware component service **130** may be managed by application **111**.

(12) Application **111** may be a specialized application (e.g., downloaded from hardware component service **130** or provided by aircraft component service **130** for download through a third-party system), or may be accessed through a browser installed on client device **110**. Application **111** may be used to enter a search query relating to a hardware component.

(13) Network **120** may be any network that transmits data communications between at least client device **110** and hardware component service **130**. Network **120** may transmit data communications between any entity shown in FIG. 1, as well as any entity discussed herein but not shown in FIG. 1. Exemplary data networks include the Internet, a wide area network, a local area network, a WiFi network, and any other network that transfers data between electronic devices.

(14) Hardware component service **130** uses both historical and new data entries relating to hardware components to generate a searchable database (e.g., input database **140**). Hardware component service **130** may be driven by a machine learning model that is trained using one or both of the historical and new data entries. Further details about hardware component service **130** are described with reference to FIG. 2 below.

(15) Input database **140** stores one or more searchable databases generated by hardware component service **130**. Hardware component service **130** may search input database **140** responsive to receiving a query relating to a given hardware component. While depicted as a separate entity from hardware component service **130**, input database **140** may be stored within the boundaries of hardware component service **130**. Source databases **150** house historical and new data entries relating to hardware components, and provide this data to hardware component service **130**.

(16) Exemplary Particulars of Hardware Component Service

(17) FIG. 2 illustrates one embodiment of exemplary modules and databases used by the hardware component service. As depicted in FIG. 2, hardware component service **130** includes historical data retrieval module **231**, weighting module **232**, training module **233**, new data module **234**, re-training module **235**, searchable database generation module **236**, search module **237**, matching module **238**, cache memory **240**, and model database **241**. The modules and databases depicted in FIG. 2 are merely exemplary; fewer or more modules and/or databases may be used to achieve the

functionality described herein. Moreover, some or all functionality of hardware component service **130** may be distributed and/or instantiated at client device **110** (e.g., on application **111**) and/or at source database **140**.

(18) Historical data retrieval module **231** retrieves historical data from any number of databases (e.g., source databases **150**). The historical data includes entries that reference both a particular airplane hardware component, and a value associated therewith. Historical data retrieval module **231** may identify source databases **150** through any known means, such as scraping web data having references to known hardware components. The references to known hardware components may be any identifying feature of a hardware component, such as a name, model number, serial number, or any other identifier. Historical data may include any data referencing any known hardware component, regardless of whether it is old data (e.g., 50 or more years old), inaccurate data, or from sources that are disreputable (e.g., based on internally calculated or third-party calculated trust scores relating to the sources).

(19) Historical data retrieval module **231** may store the retrieved historical data in cache memory **240**. Cache memory **240** may be any memory that stores readily-retrievable data, as distinguished from non-cache memory which stores data having latency that cannot be accessed in less than a given threshold amount of time. Cache memory **240** may be stored on one or more servers of hardware component service **130** and/or may be in whole or in part stored using a third-party service.

(20) Weighting module **232** determines weights to apply to values corresponding to hardware components as indicated in each entry of the historical data. In an embodiment, weights may be applied based on heuristics, where the heuristics together are referred to as a weighting model. For example, anomalies and attenuation signals may be pre-defined. The term anomaly, as used herein, may refer to an artifact within a data entry that, if found, causes the data entry to be discarded or fully discounted to a zero weight or a de minimus weight. The term attenuation signal, as used herein, may refer to an artifact within a data entry that, if found, causes the data entry to be discounted—that is, weighted to less than a normal unit of weight.

(21) Anomalies and attenuation signals may be defined by an administrator or user of the system. An amount to discount an entry having an anomaly and/or an attenuation signal, or whether to discard a data entry having an anomaly, may be pre-defined by an administrator or user of hardware component service **130**. An exemplary anomaly may be a data entry having an artifact indicative of a source that is known to be fraudulent. Another exemplary anomaly may be a cut-off age where historical data is considered to not be useful (e.g., more than 50 years old). Exemplary attenuation signals may include age (e.g., where certain age ranges of an entry relative to a present time may each be corresponded to a respective discount amount), source (e.g., where different sources have different discount amounts, or no discount amount), and so on.

(22) In an embodiment, a data entry may have more than one artifact corresponding to an anomaly or an attenuation. In such an embodiment, weighting module **232** may discount the data entry based on any or a combination of an artifact corresponding to a largest discount and/or an aggregate of all discounts corresponding to all artifacts within the data entry. In an embodiment, artifacts may be defined by a user or administrator that correspond to a positive signal. The administrator may define an amount of positive weighting that is applied where such artifacts are found in a manner opposite to that of an attenuation signal. Together, the heuristics performed by weighting module **232** may be stored in a data structure and may be collectively referred to as weighting model.

(23) In an embodiment, weighting module **232** may apply some or all of each respective data entry of the historical data into a machine-learned model. The machine-learned model may output a weight. The machine-learned weighting model may be a supervised model or an unsupervised model. Where a supervised model is used, the machine-learned model may be trained using training data having a set of one or more artifacts, where the set is paired with a label representing a weight corresponding to the training data. Thus, the machine-learned model may match each input

data entry to a weight based on the training data.

(24) Where an unsupervised model is used, the weighting model may cluster inputs for respective hardware components based on their respective values. The weighting model may determine to not discount values that correspond to clusters, but may determine to discount outlier values that do not fall into clusters. An amount of discount may vary based on distance from a given cluster. A machine-learned model used for weighting in either manner may be referred to herein as a weighting model.

(25) In an embodiment, two or more weighting models may be used. For example, weighting module **232** may determine whether any given data entry of the historical data is suitable for input into the machine-learned weighting model, where suitability may be pre-defined as having pre-defined parameters, such as a pre-defined set of artifacts. As another example, weighting module **232** may determine suitability based on whether at least a threshold number of the artifacts within a given data entry match at least a threshold number of artifacts known to be within the training data by which weighting module **232** was trained, where a sufficient match yields a determination of suitability. Where suitability is determined, weighting module **232** may apply the data entry to the machine-learned weighting model. Where suitability is determined to not exist, weighting module **232** may apply the data entry to a heuristic-driven weighting model. A technical advantage in such a hybrid weighting model system is that accuracy is maximized based on selective use of heuristics versus a machine-learned model. Moreover, applying heuristics is more processor-intensive than applying an entry to a machine-learned model, and thus reducing heuristics to scenarios where a machine-learned weighting model reduces the overall computational power required in determining weightings.

(26) Regardless of whether a machine-learned model, a heuristic model, or a hybrid model is used, in an embodiment, the weighting model may consider data entries in an aggregate form when determining weighting. For example, the weighting model may determine how many data entries of the historical data (e.g., optionally filtering out data entries having anomalies first) relate to a given hardware component. The weighting model may determine not to apply a weight (or to apply a weight of one, thus causing no change) for a value of any data entry corresponding to a hardware component that has fewer than a threshold minimum of corresponding entries in the historical data.

(27) Training module **233** trains a database generation model by using the weights determined by weighting module **232** as applied to the historical data. Training module **233** generates training data by taking an identifier of the hardware component corresponding to each entry of the historical data and pairing it with a label that matches the value indicated in the data entry. Training module **233** applies the weight to the training data, such that, for a given hardware component, the amount of weight any given training data will be given in terms of its value as a source of ground truth is discounted or augmented based on the weight applied thereto. Training module **233** trains the database generation model to take a hardware component identifier as input and to output a corresponding value using the generated training data.

(28) In an embodiment, new data module **234** receives new data also having a hardware component and a respective value. The term new data, as used herein, may refer to data entries having hardware components and respective values that are received by hardware component service **130** after the initial training is performed using the historical data. New data module **234** may continue to receive new data and may batch the new data until a predefined condition is reached. Exemplary predefined conditions may include a threshold amount of new data has been batched, a predefined amount of time has elapsed since a reference point (e.g., a first new data of a batch being received, an interval of time has passed since a last re-training, etc.), and so on. Responsive to determining that the predefined condition has been reached, new data module **234** may determine that the database generation model is to be re-trained.

(29) Re-training module **235** re-trains the database generation model by determining weights for each new data entry using weighting module **232** as applied to the new data. Re-training module

235 then generates new training data in the same manner training module **233** generated training data from the historical data entries. Re-training module **235** then trains the database generation model on the basis of the training data from all historical data (e.g., including any new data from prior re-training sessions) and on the basis of the training data from the new data. In an embodiment, the training data from all historical and new data is pooled and the database generation model is trained on the aggregate pool. Advantageously, in such an embodiment, weighting need not be re-performed on the historical data, as weights may be stored and retrieved for re-training purposes, thus reducing processing power required.

(30) In an embodiment, retrieval of historical data is selectively performed depending on whether new types of data and/or signals form part of the new data. For example, re-training module **235** may determine whether the new training data includes signals and/or data types that were not considered when the database generation model was trained. Responsive to determining that the new training data includes new data types and/or signals, re-training module **235** may generate the aggregate pool using the historical data (e.g., to ensure data relating to that type and/or signal is extracted from the historical data). However, responsive to determining that the new training data does not include new data types and/or signals, re-training module **235** may use the new data without the historical data to re-train the database generation model, modifying existing associations within the database generation model based on the new data. Similarly, where the new training data does not include new data types and/or signals, re-training module **235** may extract from the last (most recent) version of the searchable data structure values for given hardware types, and may generate training data therefrom that labels the hardware type with the value. This may be performed in place of retrieving the historical data, and may be used in conjunction with the new training data to re-train the database generation model. These manners of selectively retrieving the historical data improve on memory and bandwidth efficiency in avoiding retrieval of historical data unless it is necessary.

(31) Searchable database generation module **236** generates a searchable database of hardware components as mapped to other parameters including one or more values associated with those hardware components. Searchable database generation module **236** may generate the searchable database responsive to detecting a trigger. The term trigger, as used herein in this context, may refer to any predefined condition that causes searchable database generation module **236** to generate the searchable database. Exemplary triggers include predefined timing conditions (e.g., a threshold amount of time has passed since a last database generation and/or since a reference time (e.g., amount of time since a most recent new data has been received)), a threshold amount of new data has been received since a last generation, and so on). A trigger may also be a command manually entered by a user or an administrator. When searchable database generation module **236** generates the searchable database, a prior version may be replaced (e.g., deleted) by the newly generated version, or the prior version may be stored to memory for reference at a later time.

(32) In order to generate the searchable database, searchable database generation module **236** takes the known hardware component identifiers from a prior searchable database and inputs those into the trained database generation model. The trained database generation model outputs the hardware component identifiers as mapped to their respective values and optionally other information. The other information may include a confidence value that the respective value for a given hardware component is correct. The other information may include any other information relating to the hardware component (e.g., expected time to obtain component, expected amount of time between replacements, identification of similar components, and so on). Database generation module **236** may generate a searchable data structure from the output of the trained database generation model as indexed by hardware component.

(33) In an embodiment, responsive to detecting a trigger, database generation module **236** determines whether there is new data to be synthesized prior to generating the searchable data structure. For example, where the trigger is time-based (rather than based on new data being

detected), database generation module **236** may determine responsive to detecting the trigger whether there is new data to be synthesized. Where there is no new data, database generation module **236** may refrain from generating the searchable data structure. This has the technical advantage of improving processing power efficiency, as re-building the searchable database is not needlessly performed where no changes are to be made.

(34) Search module **237** receives a query from a user comprising an indicated hardware component. The manner in which the query is generated is described in further detail with respect to FIG. 3. Search module **237** searches the searchable database for a result matching the query, e.g., by searching for an entry in the searchable database having a hardware component identifier matching the indicated hardware component. Search module **237** outputs a result comprising a value for the indicated hardware component. Optionally, search module **237** also outputs a confidence that the value is correct. The confidence value may be determined based on a variance of values in the training data. The confidence value may be determined based on a recency, variance of values, and volume in the training data. For example, the training data may be labeled with recency values, and the model may lower confidence where recency is farther out or may lower confidence by a predefined amount corresponding to age as mapped to a depreciation of confidence.

(35) In an embodiment, the query may be a request for information. Additionally or alternatively, the query may be in connection with a request to obtain the hardware component. In such an embodiment, matching module **238** receives a second query from another user indicating the another user is in possession of the indicated hardware component. Matching module **238** determines whether a value indicated by the another user matches the value for the indicated hardware component indicated by the result. The matching need not be exact, and may be within a threshold of the hardware component indicated by the result. The threshold may be calculated based on the confidence value (e.g., based on an inverse of the confidence value). For example, where a confidence value of 60% is indicated, the threshold may be determined based by taking the inverse of the 60%, which is 40%, and defining a threshold as a value that is within 40% of the value indicated by the result. The threshold may be defined by the user seeking to obtain the hardware component. Confidence values may be expressed using any numerical representation, and use of percentages is merely exemplary here. For example, confidence values may be expressed using scores (e.g., scores from 1 to 10, where 1 shows a lowest amount of confidence and 10 shows a highest amount of confidence).

(36) Responsive to determining that the value indicated by the another user matches the value for the indicated hardware component indicated by the result, matching module **238** may execute a transaction that causes the user to obtain the indicated hardware component that the another user is in possession of. In an embodiment, responsive to determining the match, matching module **238** may first prompt either or both users to confirm, prior to executing the transaction, where the transaction is executed responsive to receiving authorization from the prompted users based on the prompts.

(37) Cache memory **240** stores data for fast access by hardware component service **130**. Fast access is distinguished from slow access, where data is stored in memory remote from hardware component service **130** (e.g., a remote server) or in slower read memory that takes longer to obtain. In an embodiment, historical data (and subsequently, new data) is initially stored in cache memory **240** when received until it is used by weighting module **232** and/or training module **233**, and then it is removed from the cache memory **240** (e.g., by deleting the data or moving it to slow access memory). Model database **241** stores models hardware component service **130** uses, including weighting models and database generation models.

(38) Alternative or additional embodiments are possible based on the modules described above. In an embodiment, historical data retrieval module **231** retrieves historical data from a plurality of source databases (e.g., source databases **150**) and new data module **234** receives new data from a

plurality of input databases (e.g., input database **140**) (e.g., again including a respective hardware component identifier and a respective associated value). Hardware component service **130** may generate a synthesized set of data by identifying a second subset of data from the historical data. Training data may be generated using the synthesized set.

(39) Hardware component module **130** may generate the synthesized set of data by segmenting the historical and new data into any number of segments. In an embodiment, a first subset of the combined new and historical data may be identified that is associated with an anomaly. Anomalies may be identified in any manner described in the foregoing. As an example, in an embodiment, hardware component module **130** may input the new data and the historical data into an unsupervised machine learning model, and may receive as output from the unsupervised machine learning model an indication of outlier data (e.g., data that is a threshold distance from any given cluster produced by a clustering model). Hardware component module **130** may assign the outlier data to be part of the first subset of data.

(40) A second subset of the combined new and historical data may be identified that is associated with an attenuation signal. Attenuation signals and entries associated with attenuation signals are subjects that are described in the foregoing and apply equally here. As an example, in an embodiment, hardware component module **130** may identify, from the historical data and the new data, stale data that is dated at least a minimum threshold amount of time from a present time, and may assign the stale data to be part of the second subset of data. In such an embodiment, weighting (e.g., performed by weighting module **232**) may weight each entry of the second subset of data based on its respective attenuation signal in a manner that is inversely proportional to a respective amount of time from a present time from a date of the respective entry. A third subset of the combined new and historical data may be identified that includes the remaining data not identified for the first or second subsets.

(41) Hardware component module **130** may update the synthesized set of data by discarding (or weighting to 0) the first subset of data from the synthesized set of data, and by weighting (e.g., using weighting module **232**) the second subset of data based on its respective attenuation signal. Where the term “discarding” is used herein, this may refer to either deleting data, or to archiving the data. The third subset of data may be not subjected to weighting. Advantageously, this saves on processing power relative to the prior-described embodiment in that weighting, which may be processing-intensive, is only applied to a subset of data, thus improving on processing, memory, and power parameters. Training module **233** may then use the synthesized set of data to train a database generation model, which may be used to generate a searchable database using searchable database generation module **235**, with which queries may be processed by search module **237** and/or matching module **238** according to the foregoing.

(42) FIG. 3 illustrates one embodiment of a user interface showing exemplary manners of searching for hardware components and receiving results. As depicted in FIG. 3, user interface **300** may include search tool **310** and/or results **320**. While depicted together, these may be shown in separate screens. Search tool **310** accepts one or more hardware component identifiers. Optionally, search tool **310** may accept additional parameters (e.g., a value, where the person submitting the query is seeking to provide a hardware component). While depicted as a drop-down menu, search tool **310** may accept a hardware component identifier in any known manner (e.g., free text, drop-down, and so on). There may be many ways to identify a hardware component, and any known mechanism may be input (e.g., scan QR code or bar code using a camera sensor, manually input a serial number, inputting a name, and so on).

(43) Results **320** may include any data corresponding to a given hardware component identifier. Hardware component source **321** is a source from which a hardware component may be obtained. Value **322** is a value corresponding to the hardware component identifier as determined using the trained database generation model. Confidence score **323** is a confidence value by the trained database generation model as determined based on variance in the training data. Any other data

corresponding to a given hardware component may be included in results **320** (e.g., availability data, lead time to acquire, etc.).

(44) Computing Machine Architecture

(45) FIG. **4** is a block diagram illustrating components of an example machine able to read instructions from a machine-readable medium and execute them in a processor (or controller). Specifically, FIG. **4** shows a diagrammatic representation of a machine in the example form of a computer system **400** within which program code (e.g., software) for causing the machine to perform any one or more of the methodologies discussed herein may be executed. The program code may be comprised of instructions **424** executable by one or more processors **402**. In alternative embodiments, the machine operates as a standalone device or may be connected (e.g., networked) to other machines. In a networked deployment, the machine may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment.

(46) The machine may be a server computer, a client computer, a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a cellular telephone, a smartphone, a web appliance, a network router, switch or bridge, or any machine capable of executing instructions **424** (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines (e.g., a database cluster and/or a server cluster) that individually or jointly execute instructions **124** to perform any one or more of the methodologies discussed herein.

(47) The example computer system **400** includes a processor **402** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a digital signal processor (DSP), one or more application specific integrated circuits (ASICs), one or more radio-frequency integrated circuits (RFICs), or any combination of these), a main memory **404**, and a static memory **406**, which are configured to communicate with each other via a bus **408**. The computer system **400** may further include visual display interface **410**. The visual interface may include a software driver that enables displaying user interfaces on a screen (or display). The visual interface may display user interfaces directly (e.g., on the screen) or indirectly on a surface, window, or the like (e.g., via a visual projection unit). For ease of discussion the visual interface may be described as a screen. The visual interface **410** may include or may interface with a touch enabled screen. The computer system **400** may also include alphanumeric input device **412** (e.g., a keyboard or touch screen keyboard), a cursor control device **414** (e.g., a mouse, a trackball, a joystick, a motion sensor, or other pointing instrument), a storage unit **416**, a signal generation device **418** (e.g., a speaker), and a network interface device **420**, which also are configured to communicate via the bus **408**.

(48) The storage unit **416** includes a machine-readable medium **422** on which is stored instructions **424** (e.g., software) embodying any one or more of the methodologies or functions described herein. The instructions **424** (e.g., software) may also reside, completely or at least partially, within the main memory **404** or within the processor **402** (e.g., within a processor's cache memory) during execution thereof by the computer system **400**, the main memory **404** and the processor **402** also constituting machine-readable media. The instructions **424** (e.g., software) may be transmitted or received over a network **426** via the network interface device **420**.

(49) While machine-readable medium **422** is shown in an example embodiment to be a single medium, the term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store instructions (e.g., instructions **424**). The term “machine-readable medium” shall also be taken to include any medium that is capable of storing instructions (e.g., instructions **424**) for execution by the machine and that cause the machine to perform any one or more of the methodologies disclosed herein. The term “machine-readable medium” includes, but not be limited to, data repositories in the form of solid-state memories, optical media, and magnetic media.

(50) Exemplary Processes for Using Hardware Component Service

(51) FIG. 5 is a flow diagram showing an exemplary process for training a machine learning model to generate a searchable database relating to hardware components. Process **500** begins with one or more processors **402** of hardware component service **130** executing one or more modules to perform operations including training **502** a machine-learned model. The training includes retrieving a plurality of entries from a database (e.g., using historical data retrieval module **231**), each entry corresponding a respective hardware component to a respective value, inputting at least a portion of data of each respective entry into a weighting model (e.g., a weighting model of model database **241**, applied using weighting module **232**), the weighting model outputting a weight to be applied to each respective value, generating a training set, the training set having training data formed by pairing each respective hardware component as a label as paired to their respective values as weighted by their respective weights, and training (e.g., using training module **233**) the machine-learned model using the training set.

(52) Hardware component service **130** receives **504** new data comprising a hardware component and a respective value (e.g., using new data module **234**) and generates **506** weighted new data (e.g., also using weighting module **232**) by inputting the new data into the weighting model, the weighting model outputting a weight to be applied to the respective value of the new data. Hardware component service re-trains **508** the machine-learned model using the training set and the weighted new data (e.g., using re-training module **235**). Responsive to detecting a trigger, hardware component service **130** generates **510**, using the machine-learned model, a searchable database (e.g., using searchable database generation module **236**). Hardware component service **130** receives **512** a query from a user comprising an indicated hardware component, and searches **514** the searchable database for a result matching the query (e.g., using search module **237**). Hardware component service **130** outputs **516** a result (e.g., result **320** of search interface **300**) comprising a value for the indicated hardware component and a confidence that the value is correct (e.g., confidence score **323**).

(53) FIG. 6 is a flow diagram showing another exemplary process for generating a searchable database. Process **600** begins with one or more processors **402** of hardware component service **130** executing one or more modules to perform operations including retrieving **602** historical data (e.g., using historical data retrieval module **231**) from a plurality of source databases (e.g., source databases **150**), the historical data comprising entries each including a hardware component identifier and an associated value. Hardware component service receives **604** new data (e.g., using new data module **234**) from a plurality of input databases (e.g., one or more input databases **140**), the new data comprising a respective hardware component identifier and a respective associated value.

(54) Hardware component service **130** generates **606** a synthesized set of data by identifying a first subset of data comprising data from the historical data and the new data that is associated with an anomaly, identifying a second subset of data from the historical data and the new data that is associated with an attenuation signal, and identifying a third subset of data from the historical data and the new data that includes data not assigned to the first subset or the second subset. Hardware component service **130** updates **608** the synthesized set of data by discarding (or archiving) the first subset of data from the synthesized set of data and weighting each entry of the second subset of data based on its respective attenuation signal (e.g., using weighting module **232**).

(55) Hardware component service **130** generates **610** a searchable database, the searchable database including each hardware component named by an entry of the synthesized set of data, along with an associated value determined based on the weighted value of the entry. Hardware component service **130** receives **612** user input of a search query, and outputs **614** search results (e.g., using user interface **300**) based on a comparison of the user input of the search query to entries of the searchable database.

(56) Additional Configuration Considerations

(57) Throughout this specification, plural instances may implement components, operations, or

structures described as a single instance. Although individual operations of one or more methods are illustrated and described as separate operations, one or more of the individual operations may be performed concurrently, and nothing requires that the operations be performed in the order illustrated. Structures and functionality presented as separate components in example configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components. These and other variations, modifications, additions, and improvements fall within the scope of the subject matter herein.

(58) Certain embodiments are described herein as including logic or a number of components, modules, or mechanisms. Modules may constitute either software modules (e.g., code embodied on a machine-readable medium or in a transmission signal) or hardware modules. A hardware module is a tangible unit capable of performing certain operations and may be configured or arranged in a certain manner. In example embodiments, one or more computer systems (e.g., a standalone, client or server computer system) or one or more hardware modules of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware module that operates to perform certain operations as described herein.

(59) In various embodiments, a hardware module may be implemented mechanically or electronically. For example, a hardware module may comprise dedicated circuitry or logic that is permanently configured (e.g., as a special-purpose processor, such as a field programmable gate array (FPGA) or an application-specific integrated circuit (ASIC)) to perform certain operations. A hardware module may also comprise programmable logic or circuitry (e.g., as encompassed within a general-purpose processor or other programmable processor) that is temporarily configured by software to perform certain operations. It will be appreciated that the decision to implement a hardware module mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations.

(60) Accordingly, the term “hardware module” should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. As used herein, “hardware-implemented module” refers to a hardware module. Considering embodiments in which hardware modules are temporarily configured (e.g., programmed), each of the hardware modules need not be configured or instantiated at any one instance in time. For example, where the hardware modules comprise a general-purpose processor configured using software, the general-purpose processor may be configured as respective different hardware modules at different times. Software may accordingly configure a processor, for example, to constitute a particular hardware module at one instance of time and to constitute a different hardware module at a different instance of time.

(61) Hardware modules can provide information to, and receive information from, other hardware modules. Accordingly, the described hardware modules may be regarded as being communicatively coupled. Where multiple of such hardware modules exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) that connect the hardware modules. In embodiments in which multiple hardware modules are configured or instantiated at different times, communications between such hardware modules may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware modules have access. For example, one hardware module may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware module may then, at a later time, access the memory device to retrieve and process the stored output. Hardware modules may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information).

(62) The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented modules that operate to perform one or more operations or functions. The modules referred to herein may, in some example embodiments, comprise processor-implemented modules.

(63) Similarly, the methods described herein may be at least partially processor-implemented. For example, at least some of the operations of a method may be performed by one or processors or processor-implemented hardware modules. The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processor or processors may be located in a single location (e.g., within a home environment, an office environment or as a server farm), while in other embodiments the processors may be distributed across a number of locations.

(64) The one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., application program interfaces (APIs).)

(65) The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the one or more processors or processor-implemented modules may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example embodiments, the one or more processors or processor-implemented modules may be distributed across a number of geographic locations.

(66) Some portions of this specification are presented in terms of algorithms or symbolic representations of operations on data stored as bits or binary digital signals within a machine memory (e.g., a computer memory). These algorithms or symbolic representations are examples of techniques used by those of ordinary skill in the data processing arts to convey the substance of their work to others skilled in the art. As used herein, an “algorithm” is a self-consistent sequence of operations or similar processing leading to a desired result. In this context, algorithms and operations involve physical manipulation of physical quantities. Typically, but not necessarily, such quantities may take the form of electrical, magnetic, or optical signals capable of being stored, accessed, transferred, combined, compared, or otherwise manipulated by a machine. It is convenient at times, principally for reasons of common usage, to refer to such signals using words such as “data,” “content,” “bits,” “values,” “elements,” “symbols,” “characters,” “terms,” “numbers,” “numerals,” or the like. These words, however, are merely convenient labels and are to be associated with appropriate physical quantities.

(67) Unless specifically stated otherwise, discussions herein using words such as “processing,” “computing,” “calculating,” “determining,” “presenting,” “displaying,” or the like may refer to actions or processes of a machine (e.g., a computer) that manipulates or transforms data represented as physical (e.g., electronic, magnetic, or optical) quantities within one or more memories (e.g., volatile memory, non-volatile memory, or a combination thereof), registers, or other machine components that receive, store, transmit, or display information.

(68) As used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

(69) Some embodiments may be described using the expression “coupled” and “connected” along with their derivatives. It should be understood that these terms are not intended as synonyms for

each other. For example, some embodiments may be described using the term “connected” to indicate that two or more elements are in direct physical or electrical contact with each other. In another example, some embodiments may be described using the term “coupled” to indicate that two or more elements are in direct physical or electrical contact. The term “coupled,” however, may also mean that two or more elements are not in direct contact with each other, but yet still cooperate or interact with each other. The embodiments are not limited in this context.

(70) As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

(71) In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the invention. This description should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

(72) Upon reading this disclosure, those of skill in the art will appreciate still additional alternative structural and functional designs for a system and a process for identifying aircraft hardware components and corresponding information through the disclosed principles herein. Thus, while particular embodiments and applications have been illustrated and described, it is to be understood that the disclosed embodiments are not limited to the precise construction and components disclosed herein. Various modifications, changes and variations, which will be apparent to those skilled in the art, may be made in the arrangement, operation and details of the method and apparatus disclosed herein without departing from the spirit and scope defined in the appended claims. While hardware components described herein are generally referred to as “aircraft” components, the term aircraft is not intended to be limiting, and hardware components for any other item may be used instead.

Claims

1. A method comprising: training a machine-learned model to, based on input of an identifier of a given hardware component, output a prediction of a given value for the given hardware component, the training comprising: retrieving historical hardware component data within a plurality of entries from a database, each entry comprising a data structure corresponding a respective hardware component to a respective value for the respective hardware component on a given date; inputting at least a portion of data of each respective entry into a weighting model, the weighting model outputting a weight to be applied to each respective value, wherein the weighting model comprises an unsupervised machine learning model that clusters inputs for respective hardware components based on their respective values and discounts outlier values that do not fall into clusters by outputting a respective weight for the each outlier value that reduces the respective value, the discounts of the outlier values proportional to their distances from one or more of the clusters, the weight determined prior to training the machine-learned model and used to discount an effect of a given training example relative to other training examples based on a magnitude of the weight; generating a training set, the training set having training data formed by pairing each respective hardware component as a label as paired to their respective values, each pair weighted by its respective weight, thereby reducing an influence of at least some of the training set on the machine-learned model during training; and training the machine-learned model using the training set; receiving new data comprising a hardware component and a respective value; generating weighted

new data by inputting the new data into the weighting model, the weighting model outputting a weight to be applied to the respective value of the new data; re-training the machine-learned model using the training set and the weighted new data; responsive to detecting a trigger, generating, using the machine-learned model, a searchable database by inputting a plurality of hardware component identifiers into the machine-learned model and receiving, as output from the machine-learned model, a corresponding current value prediction for each one of the plurality of hardware component identifiers and mapping each corresponding current value prediction and its respective hardware component identifier in an entry of the searchable database; receiving a query from a user comprising an indicated hardware component; searching the searchable database for a result matching the query; and outputting a result comprising a value for the indicated hardware component and a confidence that the value is correct.

2. The method of claim 1, further comprising, responsive to completing the training of the machine-learned model using the training set, discarding the training set from cache memory.

3. The method of claim 2, further comprising: responsive to receiving the new data, storing the new data to the cache memory; and responsive to completing the re-training of the machine-learned model, deleting the new data from the cache memory.

4. The method of claim 1, wherein the weighting model outputs a weight that does not change a respective value for a respective hardware component where the respective hardware component has fewer than a threshold minimum of corresponding entries in the database.

5. The method of claim 1, wherein the hardware component is an airplane component.

6. The method of claim 1, wherein the trigger is at least one of a user command and a detecting of a defined point in time being reached.

7. The method of claim 1, further comprising determining the confidence that the value is correct based on a variance of values used to train the machine-learned model.

8. The method of claim 1, further comprising determining the confidence that the value is correct based on a recency of values used to train the machine-learned model.

9. The method of claim 1, wherein the query is in connection with a request to obtain the hardware component, and wherein the method further comprises: receiving a second query from another user indicating the another user is in possession of the indicated hardware component; determining whether a value indicated by the another user matches the value for the indicated hardware component indicated by the result; and responsive to determining that the value indicated by the another user matches the value for the indicated hardware component indicated by the result, executing a transaction that causes the user to obtain the indicated hardware component that the another user is in possession of.

10. A non-transitory computer-readable medium comprising memory with instructions encoded thereon, the instructions, when executed, causing one or more processors to perform operations, the instructions comprising instructions to: train a machine-learned model to, based on input of an identifier of a given hardware component, output a prediction of a given value for the given hardware component, by: retrieving historical hardware component data within a plurality of entries from a database, each entry comprising a data structure corresponding a respective hardware component to a respective value for the respective hardware component on a given date; inputting at least a portion of data of each respective entry into a weighting model, the weighting model outputting a weight to be applied to each respective value, wherein the weighting model comprises an unsupervised machine learning model that clusters inputs for respective hardware components based on their respective values and discounts outlier values that do not fall into clusters by outputting a respective weight for the each outlier value that reduces the respective value, the discounts of the outlier values proportional to their distances from one or more of the clusters, the weight determined prior to training the machine-learned model and used to discount an effect of a given training example relative to other training examples based on a magnitude of the weight; generating a training set, the training set having training data formed by pairing each respective

hardware component as a label as paired to their respective values, each pair weighted by its respective weight, thereby reducing an influence of at least some of the training set on the machine-learned model during training; and training the machine-learned model using the training set; receive new data comprising a hardware component and a respective value; generate weighted new data by inputting the new data into the weighting model, the weighting model outputting a weight to be applied to the respective value of the new data; re-train the machine-learned model using the training set and the weighted new data; responsive to detecting a trigger, generate, using the machine-learned model, a searchable database by inputting a plurality of hardware component identifiers into the machine-learned model and receiving, as output from the machine-learned model, a corresponding current value prediction for each one of the plurality of hardware component identifiers and mapping each corresponding current value prediction and its respective hardware component identifier in an entry of the searchable database; receive a query from a user comprising an indicated hardware component; search the searchable database for a result matching the query; and output a result comprising a value for the indicated hardware component and a confidence that the value is correct.

11. The non-transitory computer-readable medium of claim 10, the instructions further comprising instructions to, responsive to completing the training of the machine-learned model using the training set, discard the training set from cache memory.

12. The non-transitory computer-readable medium of claim 11, the instructions further comprising instructions to: responsive to receiving the new data, store the new data to the cache memory; and responsive to completing the re-training of the machine-learned model, delete the new data from the cache memory.

13. The non-transitory computer-readable medium of claim 10, wherein the weighting model outputs a weight that does not change a respective value for a respective hardware component where the respective hardware component has fewer than a threshold minimum of corresponding entries in the database.

14. The non-transitory computer-readable medium of claim 10, wherein the hardware component is an airplane component.

15. The non-transitory computer-readable medium of claim 10, wherein the trigger is at least one of a user command and a detecting of a defined point in time being reached.

16. The non-transitory computer-readable medium of claim 10, the instructions further comprising instructions to determine the confidence that the value is correct based on a variance of values used to train the machine-learned model.

17. The non-transitory computer-readable medium of claim 10, the instructions further comprising instructions to determine the confidence that the value is correct based on a recency of values used to train the machine-learned model.

18. The non-transitory computer-readable medium of claim 10, wherein the query is in connection with a request to obtain the hardware component, and wherein the instructions further comprise instructions to: receive a second query from another user indicating the another user is in possession of the indicated hardware component; determine whether a value indicated by the another user matches the value for the indicated hardware component indicated by the result; and responsive to determining that the value indicated by the another user matches the value for the indicated hardware component indicated by the result, execute a transaction that causes the user to obtain the indicated hardware component that the another user is in possession of.
